UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 01129

LH STATISTICAL PHYSICS

SUMMER EXAMINATION 2024

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

No calculator is permitted in this examination.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. A capacitor of capacitance C and at a potential U has a Hamiltonian

$$\mathcal{H}_0(Q) = \frac{Q^2}{2C} - QU \ ,$$

where the charge on the capacitor is Q (which may be any real number).

The density of states, $W(E, C, U)$, at energy E, is expressed as (q_0 is a normalisation)

$$W(E, C, U) = \int_{-\infty}^{\infty} \frac{\mathrm{d}Q}{q_0} \, \delta\left(E - \mathcal{H}_0(Q)\right) \ .$$

(a) Show that the canonical partition function, $Z(T, C, U)$, obeys **[5]**

$$Z(T, C, U) = \int_{-\infty}^{\infty} \frac{\mathrm{d}Q}{q_0} \, \exp\left[-\beta\left(\frac{Q^2}{2C} - QU\right)\right] \ .$$

(b) Hence show that **[5]**

$$Z(T, C, U) = \sqrt{2\pi C \, \mathsf{k_B} T / q_0^2} \, \mathrm{e}^{\beta C U^2/2} \ .$$

(c) Using the expression for the Helmholtz free energy, $F(T, C, U)$ and deduce the entropy, $S(T, C, U)$. **[12]**

(d) Comment on the form of $S(T, C, U)$ as $T \to 0$. **[3]**

Now assume $U = 0$ and include a magnetic flux, Φ, and associated inductance, L, so that the Hamiltonian becomes

$$\mathcal{H}(Q, \Phi) = \frac{\Phi^2}{2L} + \frac{Q^2}{2C} \ ,$$

and the density of states (microcanonical ensemble, now normalised using Planck's constant) is

$$W(E, C, L) = \int \frac{\mathrm{d}Q \, \mathrm{d}\Phi}{\mathsf{h}} \, \delta\left(E - \frac{\Phi^2}{2L} - \frac{Q^2}{2C}\right) \ .$$

(e) Deduce a closed form for $W(E, C, L)$ (by using "polar coordinates" in the Q-Φ plane, or otherwise) and hence an expression for the temperature, $T(E, C, L)$, for $E > 0$. **[5]**

2. Consider a classical ideal gas of N identical particles with mass m residing in a **one-dimensional** region of extent L, experiencing a constant potential energy, $-U$, where $U > 0$. The canonical partition function is

$$Z(T, L, N) = \frac{1}{N!\mathsf{h}^N} \int_0^L \prod_{n=1}^{N} \mathrm{d}q_n \int_{-\infty}^{\infty} \prod_{n=1}^{N} \mathrm{d}p_n \, \exp\left[-\beta \sum_{n=1}^{N} \left(\frac{p_n^2}{2m} - U\right)\right] .$$

(a) Show the canonical partition function, $Z(T, L, N)$, obeys, for **large** N, **[10]**

$$Z(T, L, N) = \left(\frac{\mathrm{e}\ell\, \mathrm{e}^{\beta U}}{\lambda_\mathsf{T}}\right)^N .$$

where $\ell = L/N$ and $\lambda_\mathsf{T} = \left(\mathsf{h}^2/(2\pi m \mathsf{k_B} T)\right)^{1/2}$ is the thermal de Broglie wavelength.

(b) Write down the Helmholtz Free energy, $F(T, L, N)$, associated with $Z(T, L, N)$. **[2]**

(c) Remembering that $\ln \mathrm{e} = 1$, show that the chemical potential, $\mu(T, L, N)$, is **[4]**

$$\mu(T, L, N) = -\mathsf{k_B} T \, \ln\left(\frac{\ell \mathrm{e}^{\beta U}}{\lambda_\mathsf{T}}\right) .$$

The particles discussed earlier in this question, of mass m, are bound states of two particles of mass m_1 and m_2, with binding energy U and $m_1 + m_2 = m$. The remainder of this question assumes that the three "species", with average particle numbers and chemical potentials N_1 and μ_1, N_2 and μ_2 and N and μ, are in equilibrium at temperature T.

(d) What is the physical interpretation of the chemical potential, μ? Comment briefly on why is it negative in the physical case with classical particles. **[3]**

(e) Explain why the condition for equilibrium (both thermal and allowing for the bound states to beak up and recombine) among the three species is $\mu_1 + \mu_2 = \mu$. **[3]**

(f) Regarding all three species as being classical ideal gases, show that: **[4]**

$$\frac{\ell_1}{\lambda_{1\mathsf{T}}} \frac{\ell_2}{\lambda_{2\mathsf{T}}} = \frac{\ell\, \mathrm{e}^{\beta U}}{\lambda_\mathsf{T}} \text{, where for } i = 1, 2, \;\; \ell_i = \frac{L}{N_i} \quad \text{and} \quad \lambda_{i\mathsf{T}} = \left(\frac{\mathsf{h}^2}{2\pi m_i \mathsf{k_B} T}\right)^{1/2} .$$

(g) Determine the value of N as $L \to \infty$, with T held constant. **[2]**

(h) What is the physical interpretation of this result? **[2]**

3. Consider a volume V containing an ideal gas of N fermions with mass m and spin S, i.e. $2S+1$ spin states at $T=0$. The density of states, $g(\varepsilon)$ is:

$$g(\varepsilon) = V\frac{2S+1}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\varepsilon^{1/2}\,.$$

(a) Show that the Fermi energy, ε_{F} is **[4]**

$$\varepsilon_{\mathsf{F}} = \frac{\hbar^2}{2m}\left(\frac{6\pi^2 n}{2S+1}\right)^{2/3}\,,$$

where $n = N/V$ is the number density.

(b) Show that, at $T=0$, the internal energy, $U(T=0)$, is given by **[6]**

$$U(T=0) = \frac{3}{10}\left(\frac{6\pi^2}{2S+1}\right)^{2/3}\frac{\hbar^2 n^{2/3}}{m}N\,.$$

(c) Determine the pressure at $T=0$, $p(T=0)$. **[6]**

A frictionless, sliding, partition separates two compartments of a container. Compartment 1 contains spin-S_1 fermions and compartment 2 contains spin-S_2 fermions, where $S_2 > S_1$. All the particles have the same mass, m, and the temperature is $T=0$.

(d) Find the relative number density, n_1/n_2, of the two gases in equilibrium. **[6]**

(e) Determine and explain the value of the relative number density as $T\to\infty$. **[4]**

(f) Returning to $T=0$, if the gas with spin S_2 were replaced by an ideal bose gas, with mass m, deduce the relative densities in equilibrium. **[4]**

Formula Sheet

The first law of thermodynamics

$$\mathrm{d}U = T\mathrm{d}S - p\,\mathrm{d}V$$

Thermodynamic functions of state

$F = U - TS$	(Helmholtz) Free Energy
$H = U + pV$	Enthalpy
$G = F + pV$	Gibbs Free Energy

The first law of thermodynamics with varying particle number

$$\mathrm{d}U = T\mathrm{d}S - p\,\mathrm{d}V + \mu\,\mathrm{d}N$$

The grand potential

$$\Omega = F - \mu N = U - TS - \mu N\,, \qquad \mathrm{d}\Omega = -S\mathrm{d}T - p\,\mathrm{d}V - N\mathrm{d}\mu$$

The one-particle statistical weight and density of modes in d-dimensions

$$\mathrm{d}\Gamma = \frac{\mathrm{d}^d r\,\mathrm{d}^d p}{\mathsf{h}^d} \quad \text{and} \quad n_k^d = \left(\frac{L}{2\pi}\right)^d$$

Relation of canonical partition function to thermodynamic functions

$$F(T,V,N) = -k_{\mathsf{B}} T \ln Z(T,V,N) \qquad U(T,V,N) = \frac{\partial}{\partial\beta}(\beta F)$$

Gamma function and Stirling's approximation

$$\Gamma(N+1) = N! = \int_0^\infty \mathrm{d}x\, x^N\, \mathrm{e}^{-x} \quad \text{and} \quad N! \approx \left(\frac{N}{\mathrm{e}}\right)^N \Leftrightarrow \ \ln N! \approx N(\ln N - 1)\,.$$

Gaussian integral

$$\int_{-\infty}^{\infty} \mathrm{e}^{-\alpha x^2}\mathrm{d}x = \sqrt{\frac{\pi}{\alpha}}\,.$$

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

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